



David M. Everly Jr

Software Engineer – Systems, Safety & Reliability

MSN, CRNP, AGACNP-BC
<https://www.dmeverly.com>

PROFESSIONAL SUMMARY

I am a Software Engineer specializing in the design and implementation of systems emphasizing safety and reliability in high-stakes environments. My work integrates clinical-grade risk assessment and systems thinking into software and AI systems, orchestration frameworks, and workflow automation to address ambiguity and downstream risk. With over a decade of experience in critical care and emergency medicine, I bring a clinician's perspective to building robust, failure-aware systems that perform reliably under uncertainty.

SYSTEMS & TECHNICAL FOCUS

AI system orchestration · failure-aware design · LLM guardrails · applied machine learning · workflow automation · protocol design · reliability under constraint

SELECTED ENGINEERING PROJECTS

SynapSys — Failure-Aware LLM Orchestration Framework

- Designed modular orchestration layer enforcing validation, guardrails, and bounded repair
- Explicitly models failure modes, misuse, and controlled degradation around probabilistic output

Portfolio Chatbot — SynapSys Reference Implementation

- Built retrieval-augmented LLM system with pre/post guard pipelines and deterministic fallback
- Demonstrates policy-constrained interaction and trust-boundary enforcement

Additional Engineering Work

- Applied machine learning on clinical signal data (ECG classification, imbalance handling)
- Constraint-driven scheduling and workflow automation systems
- Protocol design and stateful client-server interaction (QUIC prototype)

CLINICAL & OPERATIONAL EXPERIENCE

WellSpan Pulmonology & Critical Care Medicine — Nurse Practitioner 2019–Present

- Manage high-acuity patients under time pressure, ambiguity, and evolving conditions
- Perform advanced procedures and lead real-time clinical decision-making
- Identify workflow, documentation, and system failure points impacting safety and efficiency
- Preceptor for advanced practice clinicians and trainees

Hershey Medical Center Emergency Medicine — Nurse Practitioner 2017–2019

- Emergency medicine practice in high-volume, safety-critical settings

Registered Nursing Roles 2013–2017

- Progressive care, emergency medicine, cardiopulmonary step-down, clinical instruction

EDUCATION

M.S. Computer Science (in progress) — Drexel University 2022–2026

Graduate Certificate, Computational AI — Drexel University 2024–2025

M.S. Nursing — The Pennsylvania State University 2015–2017

CREDENTIALS & RECOGNITION

ISC2 Cybersecurity Certified 2025

Epic Physician Builder 2024

AGACNP-BC 2017

Registered Nurse 2013

Advanced Practice Provider of the Year — Hershey Medical Center 2019

Graduate Fellowship — The Pennsylvania State University 2023

ENRS Abstract 2017